

JINGAN PENG

pj.peng@ou.edu | Norman, OK

EDUCATION

University of Oklahoma, Norman, OK

Ph.D. in Industrial & Systems Engineering

August 2024 - Present

- GPA: 4.0/4.0
- Advisor: Dr. Ziho Kang
- Focus on: Human Factors & Ergonomics, Neuroergonomics, Eye-tracking, Data Mining, and Virtual Reality

University of Oklahoma, Norman, OK

Data Analytics, Graduate Certificate

January 2025 – May 2026

- GPA: 4.0/4.0
- Notable Courses: Intelligent Data Analytics, Artificial Intelligence, Fundamentals of Engineering Statistical Analysis

University of Virginia, Charlottesville, VA

M.Eng. in Systems Engineering

September 2022 - December 2023

- GPA: 3.9/4.0
- Focused on: UX, UI, and user experience design, with coursework covering HCI, Human Factors, and Systems Analysis & Design.

Anhui University of Science and Technology (AUST), Huainan, China

B.Eng. in Ammunition Engineering and Explosion Technology

September 2018 - June 2022

- Junior/senior year GPA: 3.8/4.0

PUBLICATIONS

- **Peng, J., & Kang, Z.** (2026). Assessment of Cognitive Engagement in Virtual Reality Using Near-Infrared Spectroscopy (fNIRS). Proceedings of the Human Factors and Ergonomics Society (HFES) 70th International Annual Meeting (ASPIRE), Reno, NV. (Accepted)
- **Peng, J., Kang, Z., & Park, B. B.** (2026). Stochastic simulation of an adaptive cruise control (ACC) system in electric vehicles to investigate human perceptions of safety, consistency, and comfort. *Traffic Injury Prevention*, 1-10.
- **Peng, J., Kang, Z., Lee, J., & Fraga, R. P.** (2025, May). Evaluation of Brain Signal Quality in Virtual Reality: Detection of Infrared Signal Interference Using the fNIRS Device. In 2025 6th International Conference on Bio-engineering for Smart Technologies (BioSMART) (pp. 1-4). IEEE.
- Camacho, G., Bolton, M., **Peng, J.**, Wagle, P., & Feng, L. (2023). A Formal Method for the Analysis of the Veteran's Ebenefits' Website. *Human Factors in Software and Systems Engineering*, 94(94).

RESEARCH EXPERIENCE

Human Engagement Exploring in Multi-Person Virtual Reality (VR), Norman, OK

August 2024 - Present

- Conducted multi-person VR experiments to collect biometric (fNIRS, eye-tracking) and behavioral data for assessing learning engagement and assessment in virtual environments.
- Contributed to developing the immersive VR experiment scenarios using Vizard and Godot platforms.
- Led end-to-end fNIRS data analysis, including preprocessing, filtering, and motion artifact correction. **Identified** significant infrared interference from VR headsets and **developed** a data-driven filtering approach that effectively resolved the issue. -One first-author paper published; several follow-up manuscripts are in progress.

Human Factors Study on Automated Cruise Control (ACC) Simulation | Project Lead, Norman, OK

March 2024 - Present

- Modeled ACC behavior using MATLAB and designed scenarios simulating emergency braking and congested traffic conditions.
- Introduced human-perception-based metrics to connect simulation data with driver experience.
- Demonstrated that electric vehicles offer smoother and safer control, providing new insight into human trust and comfort in automation. - First-author paper accepted by *Traffic Injury Prevention*.

Optimization of Ebenefits/VA.gov Website Login Process, Charlottesville, VA

March 2023 - May 2023

- Contributed to the design and formulation of a probabilistic model analyzing login failures on the Ebenefits/VA.gov website.
- Implemented and refined model algorithms to ensure computational accuracy and reliability.
- Collaborated to demonstrate the effectiveness of the Defense Self-Service Logon Path in enhancing login success rates.

- National College Student Innovation Project | Project Lead**, Huainan, China

September 2019 - May 2021
- Led a five-member research team focused on optimizing the structure and materials of a new explosive formulation, and identified that graphite coating enhances both its physical and chemical stability.
 - Analyzed and visualized the thermal decomposition process using a differential scanning calorimeter and an electron microscope.
 - Earned an “excellent” evaluation for the project presentation and secured ¥20,000 in research funding.

WORK EXPERIENCE

- Human Factors & Simulation Laboratory at University of Oklahoma**, Norman, OK

Research Assistant

March 2024 - Present
- Deeply involved in multiple human factors research projects, including VR-based cognitive experiments and driving simulation studies.
 - Responsible for processing and analyzing functional near-infrared spectroscopy (fNIRS), eye-tracking, and behavioral data using Python, R, and MATLAB, including cleaning, preprocessing, modeling, and visualization.
 - Writing research reports and academic manuscripts for conference and journal submission.

- School of Industrial and Systems Engineering at University of Oklahoma**, Norman, OK

Teaching Assistant

August 2024 - Present
- Assisted in teaching undergraduate and graduate courses, including Ergonomics in Systems Design, Systems Analysis Using Simulation, and Data-Driven Decision Making.
 - Supported classroom instruction, led lab sessions, and guided semester projects, including experimental design, data analysis, and simulation modeling.
 - Graded assignments, provided feedback, and helped students understand applications of human factors principles and data-driven decision tools.

- iFLYTEK Co., Ltd**, Hefei, China

Junior Technical Support Engineer Intern

July 2021 - September 2021
- Analyzed behavior and feedback data from 1,000+ users, identifying key usability pain points and contributing to design improvements in intelligent education platforms.
 - Maintained and optimized user databases using Navicat and Fiddler to support large-scale data integration.
 - Assisted in product and user interface evaluations.

TECHNICAL SKILLS

Programming languages: R, Python, MATLAB

Machine learning: R (caret, forecast, train), Python (scikit-learn, gurobi, gymnasium)

Other Tools: Figma, Arena, SAS, PTV Vistro, Origin

AWARDS

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| • Tomorrow’s Engineers Scholarship | 2026 |
| • Frances E. & A. Earl Ziegler Sooner Heritage Scholarship | 2026 |
| • Director’s Graduate Scholarship, Gallogly College of Engineering | 2026 |
| • Thomas I. Brown, Jr. Endowed Scholarship, Gallogly College of Engineering | 2025 |
| • Ph.D. Recruitment Excellence Fellowship , Gallogly College of Engineering | 2024 |
| • Excellent Student Award & Student Scholarship, Anhui University of Science and Technology | 2019, 2020, 2021 |
| • Bronze Award, the 6th Internet + College Student Innovation and Entrepreneurship Competition | 2020 |